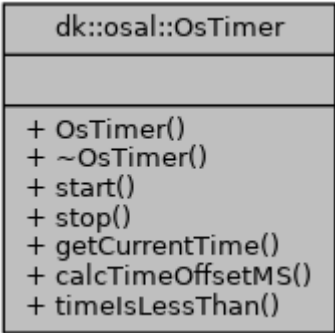


dk::osal::OsTimer Class Reference

```
#include <DKOsaiOsTimer.h>
```

Collaboration diagram for dk::osal::OsTimer:



Public Types

typedef struct timespec **Timestamp**

Public Member Functions

- OsTimer** ()
OsaiOsTimer shall abstract the control/configure timer functions provided by OS(constructor) [More...](#)
- ~OsTimer** ()
OsaiOsTimer shall abstract the control/configure timer functions provided by OS(destructor) [More...](#)
- void **start** (const **Timestamp** &expireTime)
OsaiOsTimer component shall provide APIs to start timer with expiration time. [More...](#)
- void **stop** ()
- void **getCurrentTime** (**Timestamp** &ts)
OsaiOsTimer component shall provide APIs to get the current time by using OS provided timer functions.
[More...](#)

Static Public Member Functions

- static void **calcTimeOffsetMS** (uint32_t timeoutMS, **Timestamp** &ts)
OsaiOsTimer component shall provide APIs to calculate the millisecond offset by adding the timeout with known timespec value. [More...](#)
- static bool **timeIsLessThan** (const **Timestamp** &ts1, const **Timestamp** &ts2)
OsaiOsTimer component shall provide APIs to compare the timespec values obtained for different instances. [More...](#)

Member Typedef Documentation

◆ Timestamp

```
typedef struct timespec dk::osal::OsTimer::Timestamp
```

Constructor & Destructor Documentation

◆ OsTimer()

```
dk::osal::OsTimer::OsTimer ( )
```

OsalOsTimer shall abstract the control/configure timer functions provided by OS(constructor)

requirement ID 751535

The timer_create() function creates a per-process timer using the specified clock source, clock_id, as the timing base. The timer is created in the disabled state, and isn't enabled until you call timer_settime()

Returns

: 0 Success, -1 The timerid argument is set to the timer's ID. An error occurred (errno is set).

Parameters

errors EAGAIN All timers are in use. You'll have to wait for a process to release one. EINVAL The clock_id isn't one of the valid CLOCK_* constants. EPERM The calling process doesn't have the required permission; see procmgr_ability().

requirement ID 751566

◆ ~OsTimer()

dk::osal::OsTimer::~OsTimer ()

OsalOsTimer shall abstract the control/configure timer functions provided by OS(destructor)

requirement ID 751535

The timer_delete() function removes a previously attached timer based upon the timerid returned from the timer_create() function.

The timer is removed from the active system timer list, and returned to the free list of available timers.

Returns

0 Success, -1 An error occurred (errno is set).

Parameters

errors EINVAL The timer timerid isn't attached to the calling process.

requirement ID 751567

Member Function Documentation

◆ calcTimeOffsetMS()

```
void dk::osal::OsTimer::calcTimeOffsetMS ( uint32_t      timeoutMS,  
                                             Timestamp & ts  
                                             )
```

static

OsalOsTimer component shall provide APIs to calculate the millisecond offset by adding the timeout with known timespec value.

requirement ID 751571

Here is the caller graph for this function:



◆ getCurrentTime()

```
void dk::osal::OsTimer::getCurrentTime ( Timestamp & ts )
```

OsalOsTimer component shall provide APIs to get the current time by using OS provided timer functions.

requirement ID 751570

◆ start()

```
void dk::osal::OsTimer::start ( const Timestamp & expireTime )
```

OsalOsTimer component shall provide APIs to start timer with expiration time.

OsalOsTimer component shall provide APIs to stop the timer which is already running.

requirement ID 751568

requirement ID 751569

The timer_settime() function sets the expiration time of the timer specified by timerid from the it_value member of the value argument. If the it_value structure member of value is zero, then the timer is disarmed.

Returns

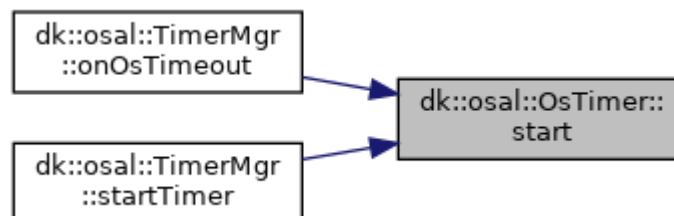
0 Success. -1 An error occurred (errno is set).

Parameters

Errors EFAULT A fault occurred trying to access the buffers provided. EINVAL The timer timerid isn't attached to the calling process or the number of nanoseconds specified by the tv_nsec member of one of the timespec structures in the itimerspec structure pointed to by value is less than zero or greater than or equal to 1000 million.

requirement ID 751568

Here is the caller graph for this function:



◆ stop()

```
void dk::osal::OsTimer::stop ( )
```

The timer_settime() function sets the expiration time of the timer specified by timerid from the it_value member of the value argument. If the it_value structure member of value is zero, then the timer is disarmed.

Returns

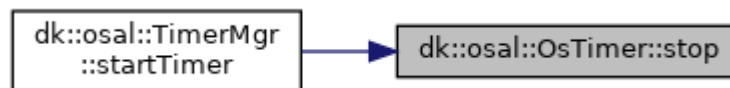
0 Success. -1 An error occurred (errno is set).

Parameters

Errors EFAULT A fault occurred trying to access the buffers provided. EINVAL The timer timerid isn't attached to the calling process or the number of nanoseconds specified by the tv_nsec member of one of the timespec structures in the itimerspec structure pointed to by value is less than zero or greater than or equal to 1000 million.

requirement ID 751569

Here is the caller graph for this function:



◆ timelsLessThan()

```
bool dk::osal::OsTimer::timelsLessThan ( const Timestamp & ts1,  
                                         const Timestamp & ts2  
                                         )
```

static

OsalOsTimer component shall provide APIs to compare the timespec values obtained for different instances.

requirement ID 751572

Here is the caller graph for this function:



The documentation for this class was generated from the following files:

- /home/damudhar/29_8_audi/cluster-platform/dijkstra/abstraction/osal/src/posix/DKOsalOsTimer.h
- /home/damudhar/29_8_audi/cluster-platform/dijkstra/abstraction/osal/src/posix/DKOsalOsTimer.cpp